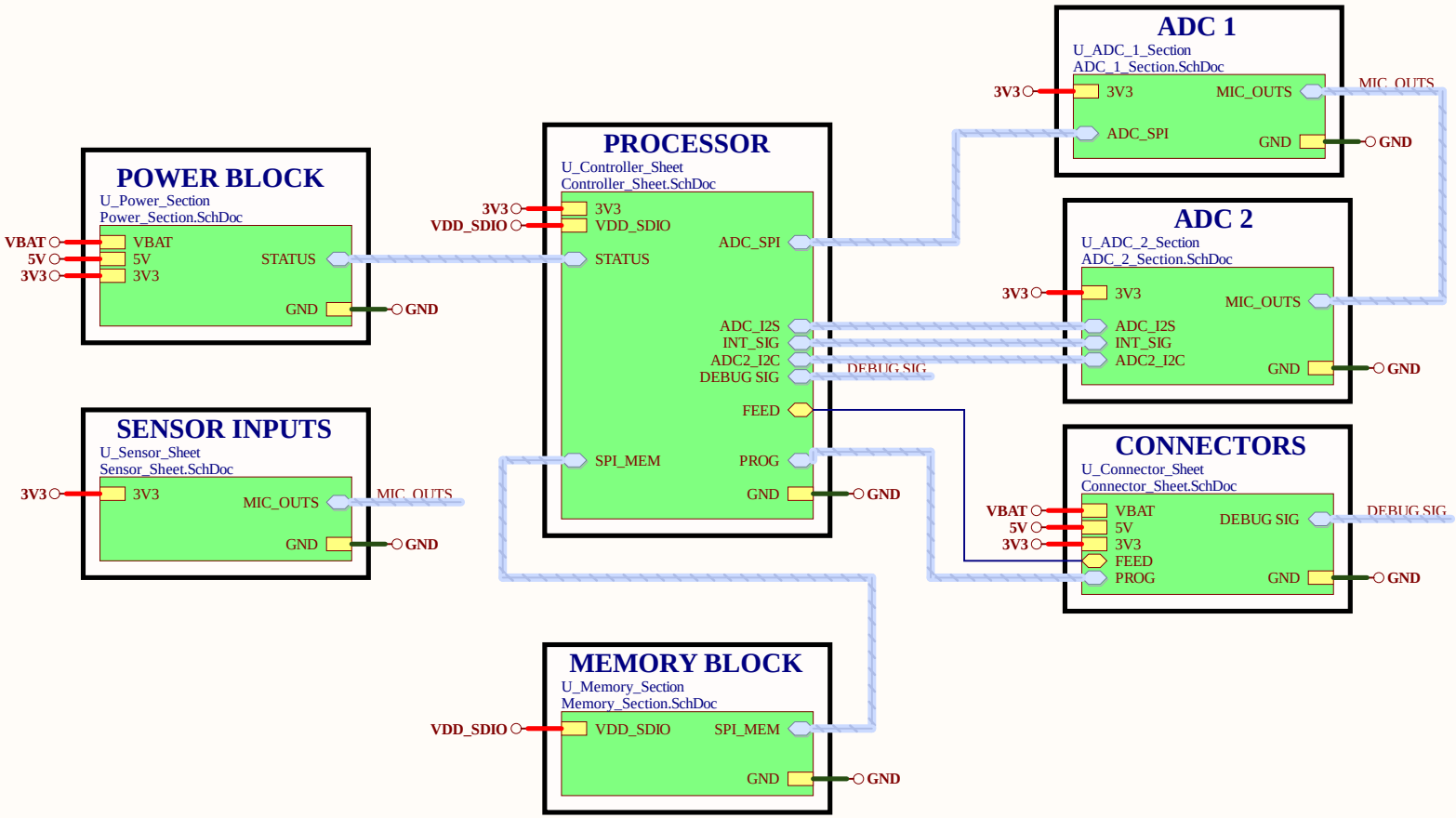
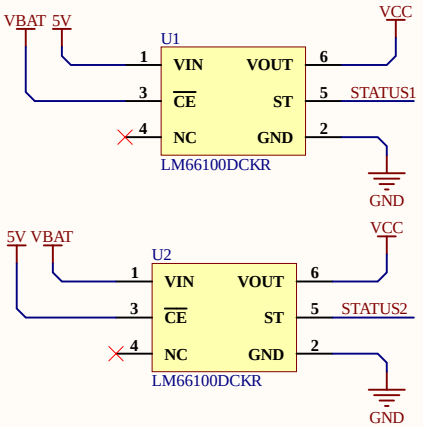


Cannot open file C:\Users\user\Pictures\Saved Pictures\espark-consulting-group-150x150.jpg. File does not exist.	Title: Sound_Sensor.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 7/14/2022	Sheet 1 of 8	
	File: E:\Melih-SoundSensor\..\Top_Sheet.SchDoc	Drawn By: Huzaifa Ali	

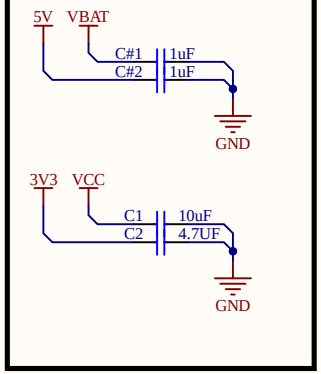
ORING STATUS SIGNAL



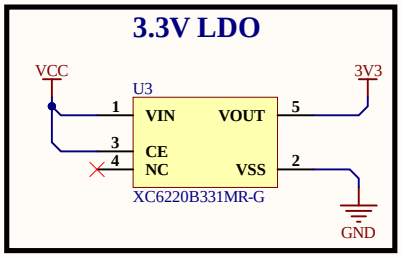
ORING CIRCUITRY



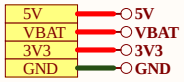
DECOUPLING CAPS



3.3V LDO

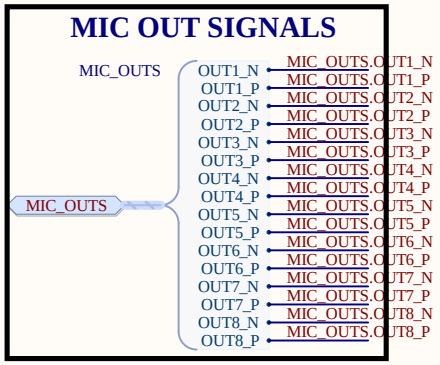


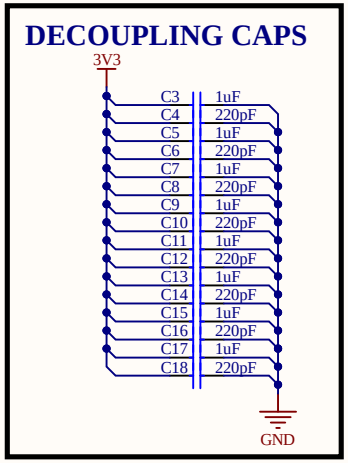
Fixed 3.3V output
150mV dropout @500mA
Max output current is 1A
Enable High to activitate the chip

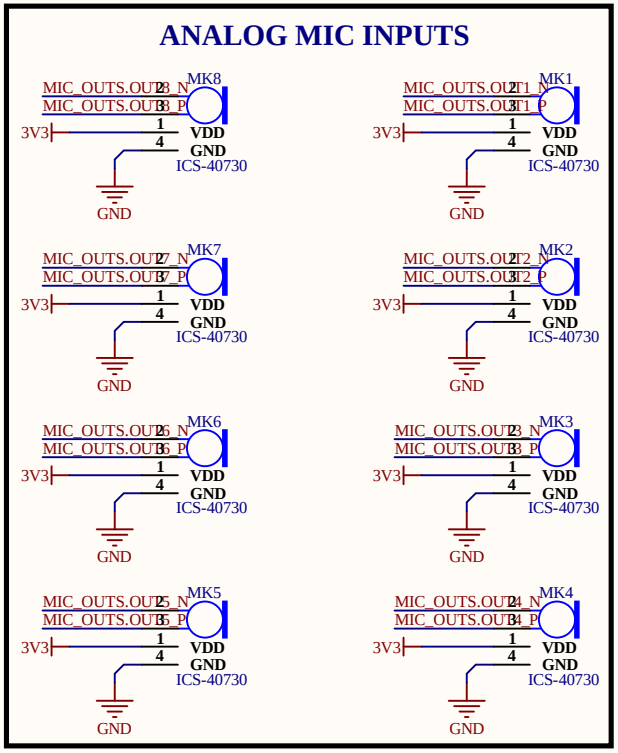


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Title: Sound_Sensor.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 7/14/2022		Sheet 2 of 8	
File: E:\Melih-SoundSensor\..\Power_Section.SchDoc		Drawn By: Huzaifa Ali	

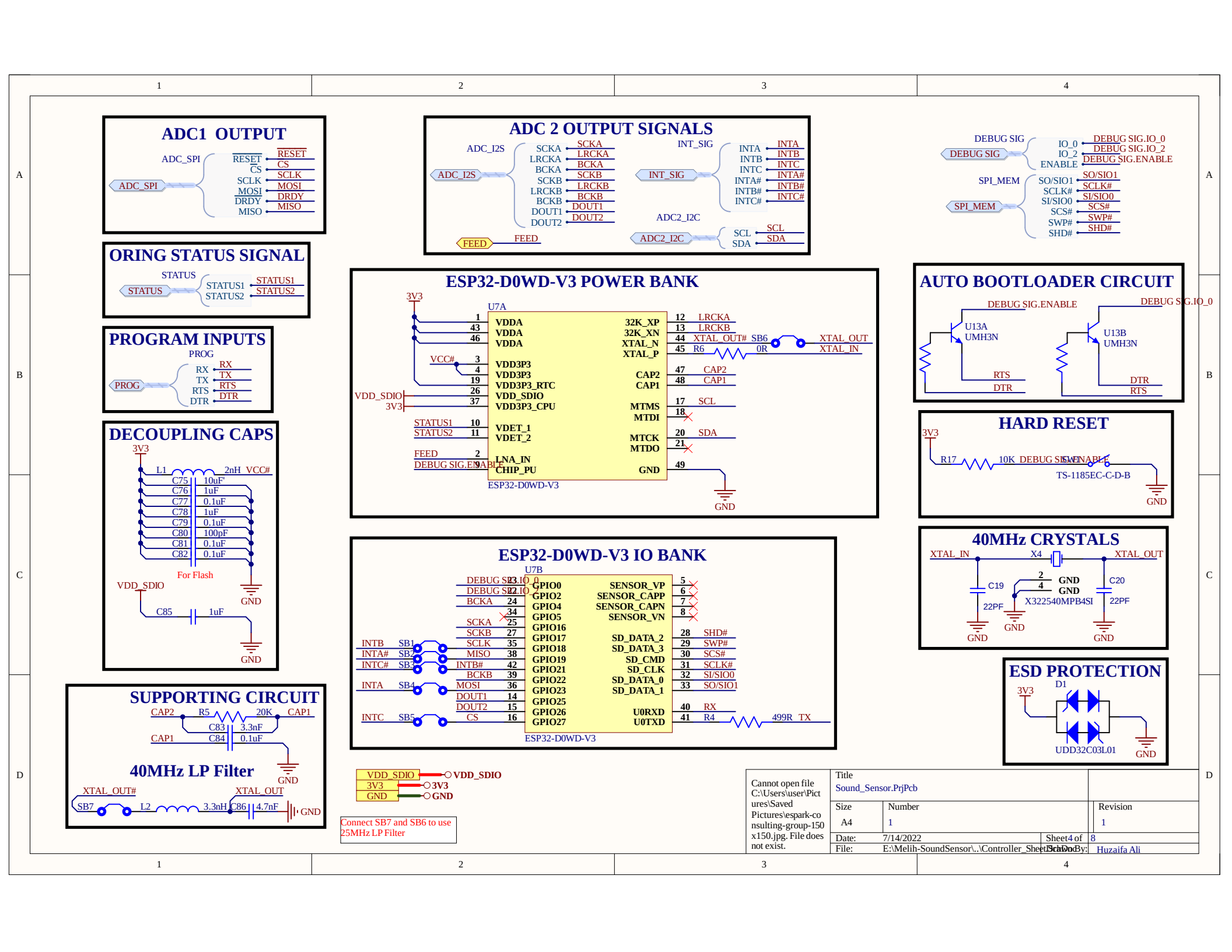


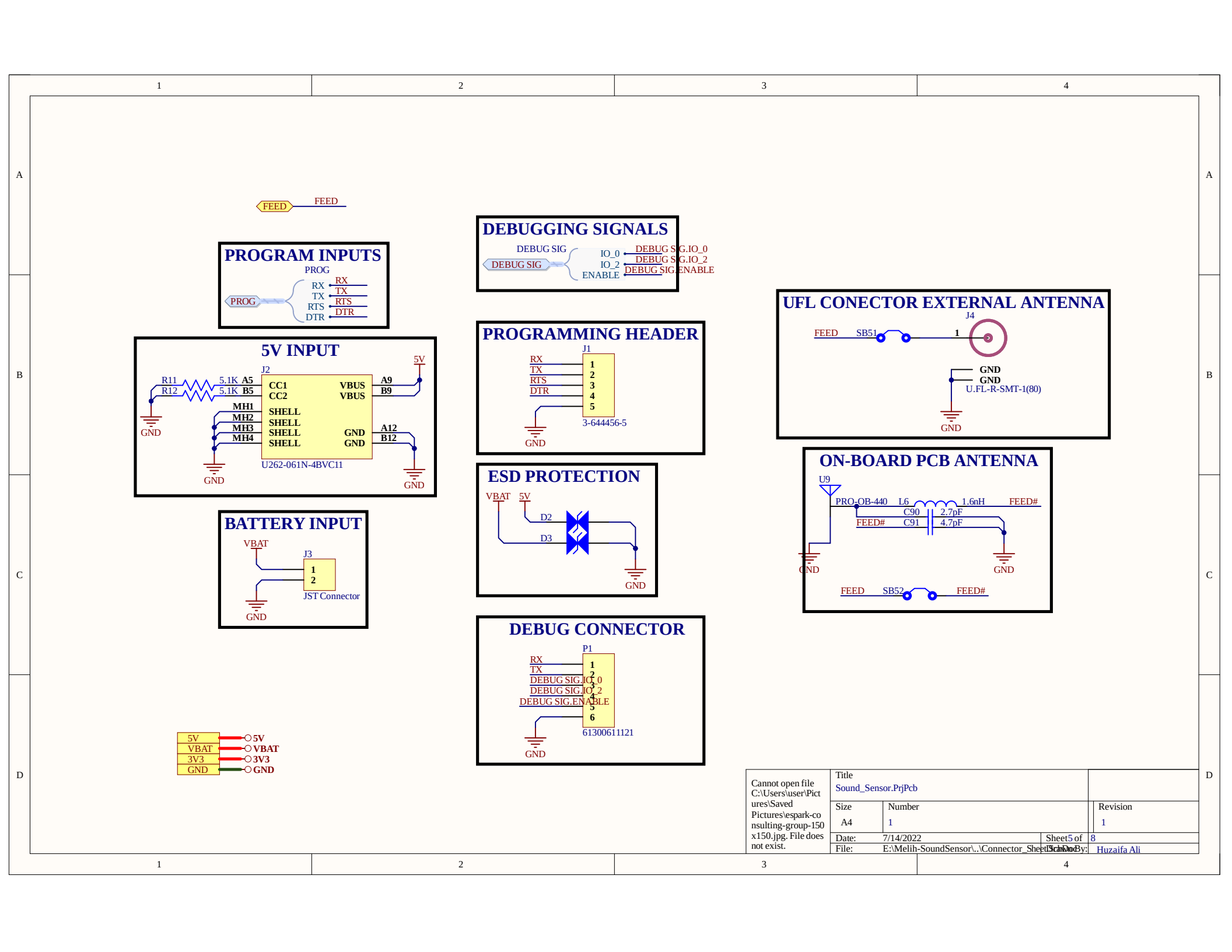




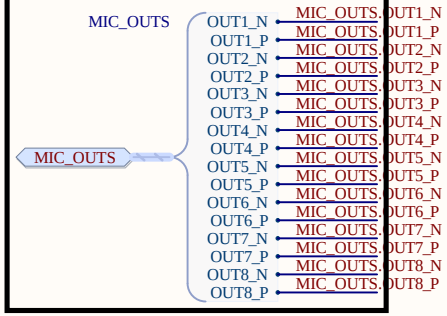


Cannot open file C:\Users\user\Pictures\Saved Pictures\espark-consulting-group-150x150.jpg. File does not exist.	Title: Sound_Sensor.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 7/14/2022	Sheet 3 of 8	
	File: E:\Melih-SoundSensor\..\Sensor_Sheet.SchDoc	Drawn By: Huzaifa Ali	

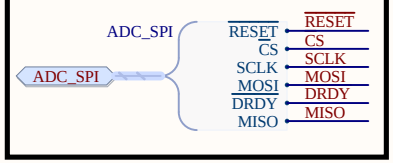

ADC1 OUTPUT
ADC_SPI
RESET
CS
SCLK
MOSI
DRDY
MISO
ADC 2 OUTPUT SIGNALS
ADC_I2S
SCKA
LRCKA
BCKA
SCKB
LRCKB
BCKB
DOUT1
DOUT2
FEED
INT_SIG
INTA
INTB
INTC
INTA#
INTB#
INTC#
ADC2_I2C
SCL
SDA
DEBUG SIG
DEBUG SIG.IO_0
DEBUG SIG.IO_2
DEBUG SIG.ENABLE
SPI_MEM
SO/SIO1
SCLK#
SI/SIO0
SCS#
SWP#
SHD#
ORING STATUS SIGNAL
STATUS
STATUS1
STATUS2
ESP32-D0WD-V3 POWER BANK
U7A
3V3
VDDA
VDD3P3
VDD3P3_RTC
VDD_SDIO
VDD3P3_CPU
VDET_1
VDET_2
LNA_IN
CHIP_PU
32K_XP
32K_XN
XTAL_N
XTAL_P
CAP2
CAP1
MTMS
MTDI
MTCK
MTDO
GND
ESP32-D0WD-V3
AUTO BOOTLOADER CIRCUIT
U13A
UMH3N
U13B
UMH3N
RTS
DTR
PROGRAM INPUTS
PROG
RX
TX
RTS
DTR
HARD RESET
R17
10K
TS-1185EC-C-D-B
DECOUPLING CAPS
L1
2nH
VCC#
C75
10uF
C76
1uF
C77
0.1uF
C78
1uF
C79
0.1uF
C80
100pF
C81
0.1uF
C82
0.1uF
For Flash
C85
1uF
ESP32-D0WD-V3 IO BANK
U7B
GPIO0
GPIO2
GPIO4
GPIO5
GPIO16
GPIO17
GPIO18
GPIO19
GPIO21
GPIO22
GPIO23
GPIO25
GPIO26
GPIO27
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
SD_DATA_2
SD_DATA_3
SD_CMD
SD_CLK
SD_DATA_0
SD_DATA_1
U0RXD
U0TXD
40MHz CRYSTALS
XTAL_IN
X4
XTAL_OUT
C19
22PF
C20
22PF
X322540MPB4SI
SUPPORTING CIRCUIT
CAP2
R5
20K
CAP1
C83
3.3nF
C84
0.1uF
40MHz LP Filter
XTAL_OUT#
XTAL_OUT
SB7
L2
3.3nH
C86
4.7nF
ESD PROTECTION
D1
UDD32C03L01
Connect SB7 and SB6 to use 25MHz LP Filter
Cannot open file C:\Users\user\Pictures\Saved Pictures\espark-consulting-group-150x150.jpg. File does not exist.
Title
Sound_Sensor.PrjPcb
Size
A4
Number
1
Revision
1
Date:
7/14/2022
Sheet4 of 8
File:
E:\Melih-SoundSensor\..\Controller_Sheet.SchDoc
Drawn By:
Huzaifa Ali


FEED
FEED
PROGRAM INPUTS
PROG
RX
TX
RTS
DTR
5V INPUT
J2
R11
R12
5.1K A5
5.1K B5
CC1
CC2
SHELL
SHELL
SHELL
SHELL
VBUS
VBUS
GND
GND
A9
B9
A12
B12
MH1
MH2
MH3
MH4
5V
GND
GND
GND
U262-061N-4BVC11
BATTERY INPUT
VBAT
J3
JST Connector
GND
5V
VBAT
3V3
GND
5V
VBAT
3V3
GND
DEBUGGING SIGNALS
DEBUG SIG
IO_0
IO_2
ENABLE
DEBUG SIG.IO_0
DEBUG SIG.IO_2
DEBUG SIG.ENABLE
PROGRAMMING HEADER
J1
RX
TX
RTS
DTR
3-644456-5
GND
ESD PROTECTION
VBAT
5V
D2
D3
GND
DEBUG CONNECTOR
P1
RX
TX
DEBUG SIG.IO_0
DEBUG SIG.IO_2
DEBUG SIG.ENABLE
61300611121
GND
UFL CONECTOR EXTERNAL ANTENNA
J4
FEED
SB51
GND
GND
U.FL-R-SMT-1(80)
GND
ON-BOARD PCB ANTENNA
U9
PRO-OB-440
L6
1.6nH
FEED#
C90
2.7pF
FEED#
C91
4.7pF
GND
GND
FEED
SB52
FEED#
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Title
Sound_Sensor.PrjPcb
Size
A4
Number
1
Revision
1
Date:
7/14/2022
Sheet5 of 8
File:
E:\Melih-SoundSensor\..\Connector_Sheet.SchDoc
Drawn By:
Huzaifa Ali

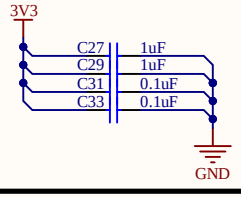
MIC OUT SIGNALS



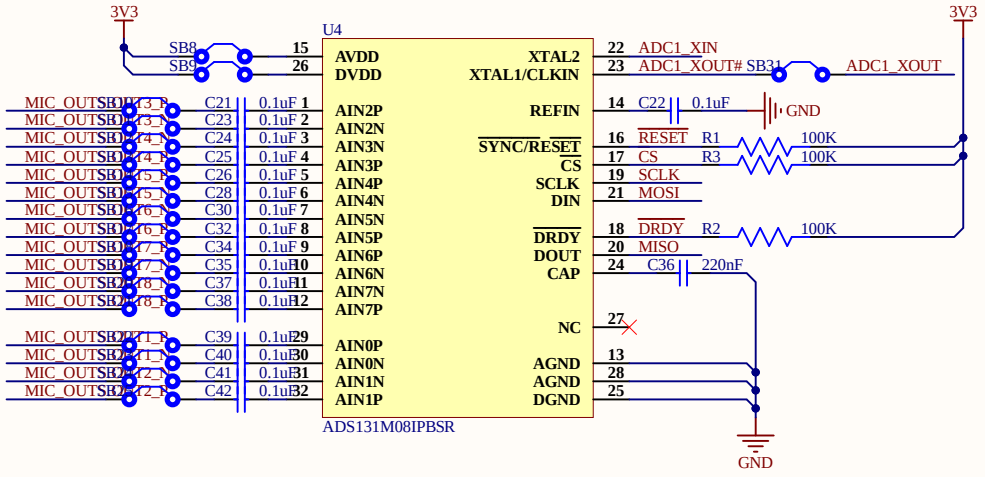
ADC OUTPUT



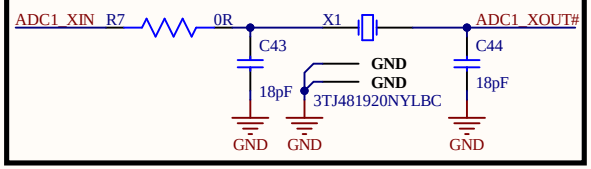
DECOUPLING CAPS



ADC WITH SPI INTERFACE

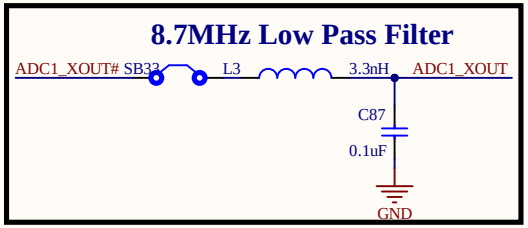


8.192 MHz Crystal



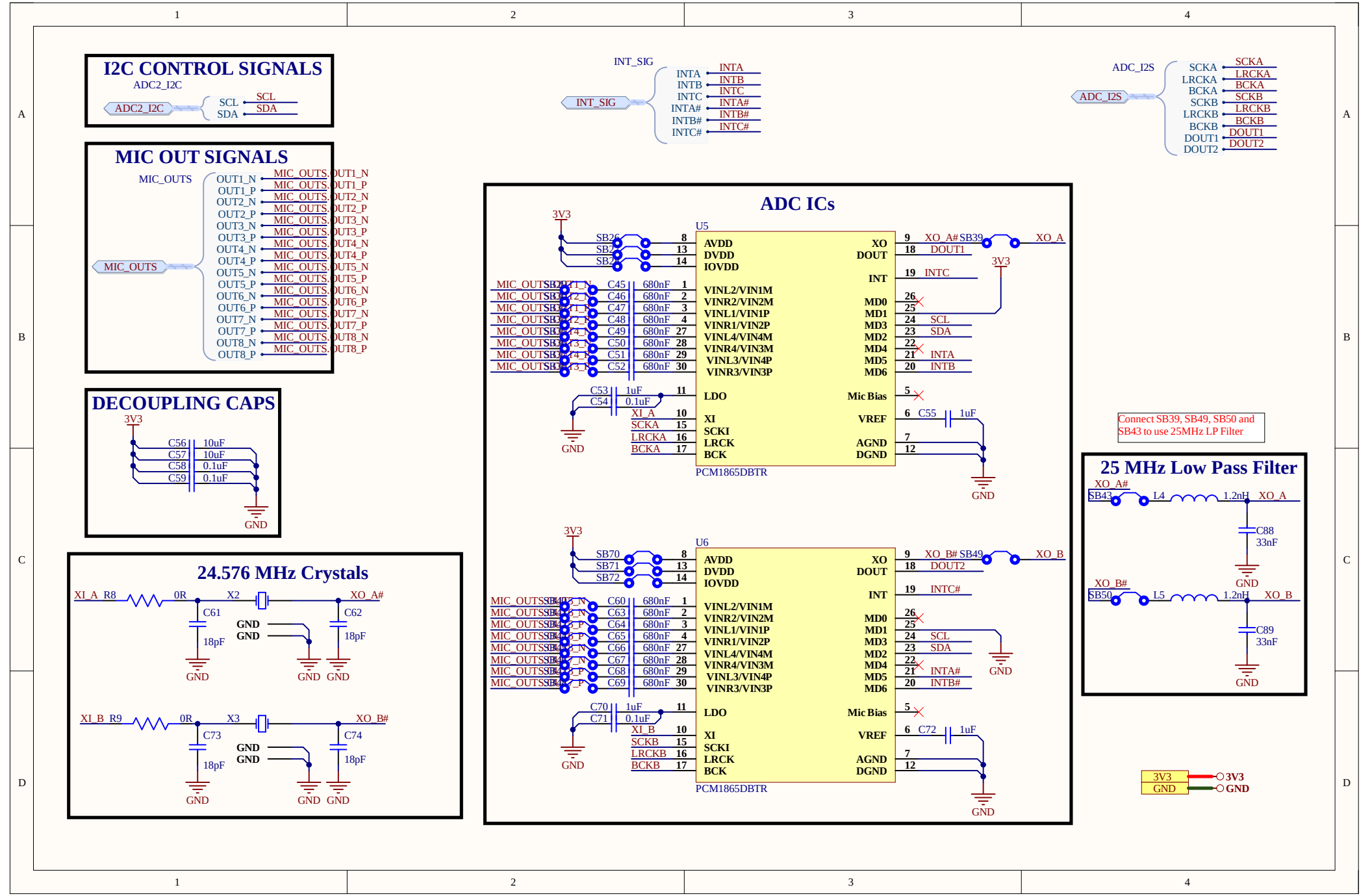


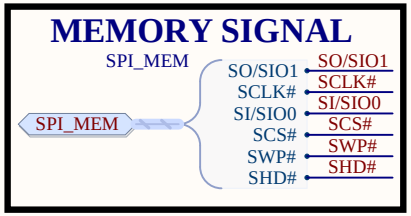
8.7MHz Low Pass Filter

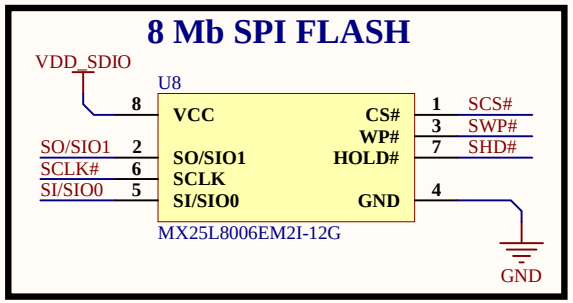


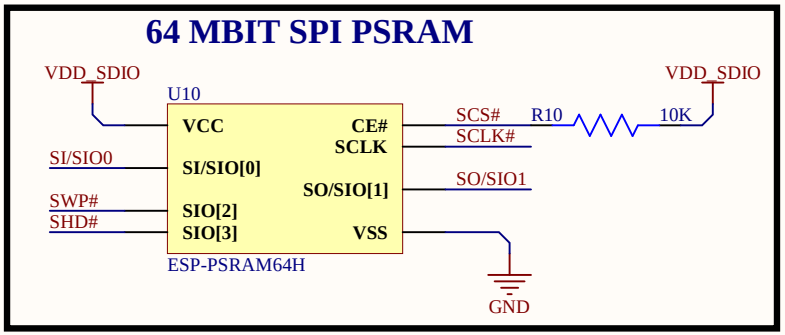
Connect SB39 and SB31 to use 25MHz LP Filter

Title		
Size A4	Number	Revision
Date: 7/14/2022		Sheet of
File: E:\Melih-SoundSensor\..\ADC_1_Section.SchDoc		Drawn By:


I2C CONTROL SIGNALS
ADC2_I2C
ADC2_I2C
SCL
SDA
SCL
SDA
MIC OUT SIGNALS
MIC_OUTS
MIC_OUTS
OUT1_N
OUT1_P
OUT2_N
OUT2_P
OUT3_N
OUT3_P
OUT4_N
OUT4_P
OUT5_N
OUT5_P
OUT6_N
OUT6_P
OUT7_N
OUT7_P
OUT8_N
OUT8_P
DECOUPLING CAPS
3V3
C56 10uF
C57 10uF
C58 0.1uF
C59 0.1uF
GND
24.576 MHz Crystals
XI_A R8 0R X2 XO_A#
C61 18pF
C62 18pF
GND GND GND GND
XI_B R9 0R X3 XO_B#
C73 18pF
C74 18pF
GND GND GND GND
INT_SIG
INT_SIG
INTA
INTB
INTC
INTA#
INTB#
INTC#
ADC_I2S
ADC_I2S
SCKA
LRCKA
BCKA
SCKB
LRCKB
BCKB
DOUT1
DOUT2
ADC ICs
U5
3V3
AVDD
DVDD
IOVDD
VINL2/VIN1M
VINR2/VIN2M
VINL1/VIN1P
VINR1/VIN2P
VINL4/VIN4M
VINR4/VIN3M
VINL3/VIN4P
VINR3/VIN3P
LDO
XI
SCKI
LRCK
BCK
XO
DOUT
INT
MD0
MD1
MD3
MD2
MD4
MD5
MD6
Mic Bias
VREF
AGND
DGND
XO_A#
XO_A
DOUT1
INTC
SCL
SDA
INTA
INTB
XI_A
SCKA
LRCKA
BCKA
C53 1uF
C54 0.1uF
C55 1uF
PCM1865DBTR
GND
U6
XO_B#
XO_B
DOUT2
INTC#
INTA#
INTB#
XI_B
SCKB
LRCKB
BCKB
C70 1uF
C71 0.1uF
C72 1uF
PCM1865DBTR
Connect SB39, SB49, SB50 and SB43 to use 25MHz LP Filter
25 MHz Low Pass Filter
XO_A#
SB43
L4 1.2nH
XO_A
C88 33nF
XO_B#
SB50
L5 1.2nH
XO_B
C89 33nF
3V3
GND
3V3
GND

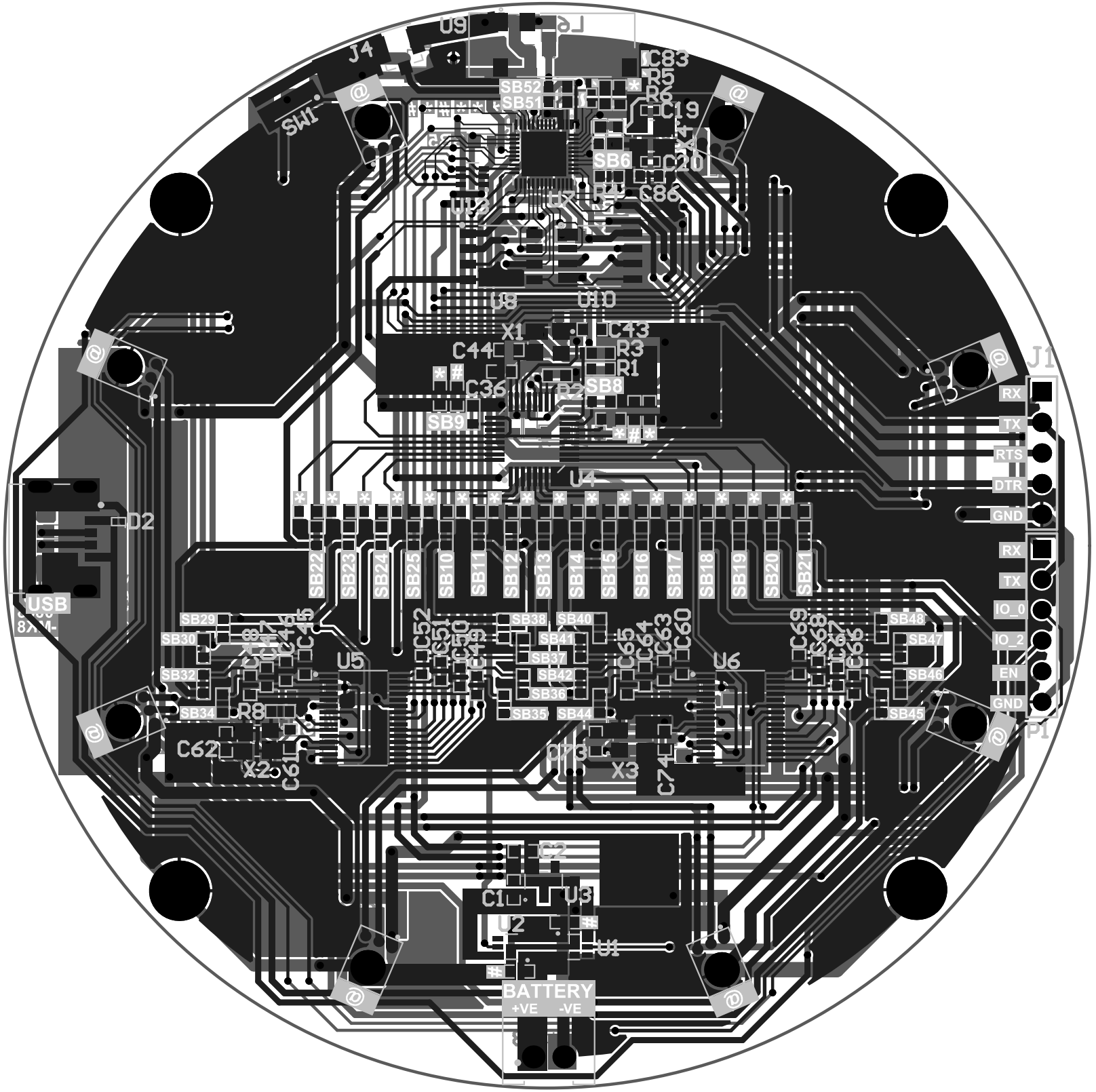








Title		
Size: A	Number:	Revision:
Date: 7/14/2022		Sheet of
File: E:\Melih-SoundSensor\..\Memory_Section.SchDoc		Drawn By:


J4
SW1
U9
IC83
SB52
SB51
R5
R6
C19
X4
SB6
C20
C86
U8
U10
X1
C43
C44
R3
R1
C36
R2
SB8
SB9
U4
J1
RX
TX
RTS
DTR
GND
RX
TX
IO_0
IO_2
EN
GND
PI
D2
USB
SB22
SB23
SB24
SB25
SB10
SB11
SB12
SB13
SB14
SB15
SB16
SB17
SB18
SB19
SB20
SB21
SB29
SB30
SB32
SB34
SB38
SB40
SB41
SB37
SB42
SB36
SB35
SB44
SB48
SB47
SB46
SB45
C52
C51
C50
C49
C65
C64
C63
C60
C69
C68
C67
C66
U5
U6
R8
C62
C61
X2
X3
C74
C2
C1
U3
U2
U1
BATTERY
+VE
-VE